

Aman Kumar

Dirac Postdoctoral Fellow
Theoretical Condensed Matter Physics
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Education

- 2023–present **Dirac Postdoctoral Fellow, Physics**
National High Magnetic Field Laboratory, Tallahassee, Florida
ADVISOR: Prof. Hitesh Changlani
- 2018–2023 **Ph.D. in Physics**
Tata Institute of Fundamental Research, Mumbai, India
ADVISOR: Prof. Vikram Tripathi
- 2016–2018 **M.Sc. in Physics**
Tata Institute of Fundamental Research, Mumbai, India
- 2013–2016 **B.Sc. (Hons.) in Physics**
University of Delhi, India

Academic Research Visit

- Sept 1st - Nov 1st 2022 **Visit to Prof. Subir Sachdev's group**
Harvard University, Cambridge, USA

Awards and Fellowships

- Sept 2023 **Dirac Fellowship**
National High Magnetic Field Laboratory
- Mar 2022 **Sarojini Damodaran International Travel Fellowship**
Tata Institute of Fundamental Research, Mumbai, India
- Mar 2022 **APS Distinguished Student Award**
- Mar 2016 **Academic Merit Award in B.Sc.**
A.R.S.D. College, University of Delhi

Areas of Research

- Wigner crystal physics in triangular moiré systems
- Study of emergent phenomena such as fractionalization and topological order
- Quasiparticle stability in quantum many-body systems such as frustrated magnets (Kitaev systems) and underdoped cuprates
- Formulation of thermal Hall transport using tensor-network methods
- Many-body localization in Fock space

Skills

- Programming: Python, C, C++, Bash
- High-performance computing
- Tensor-network techniques
- Large-scale exact diagonalization, FEAST method

- Familiarity with many-body field theory techniques

Teaching Assistant Experience

	Courses	Instructor
Jan-July. 2020	Statistical Physics I	Kedar Damle
Sept-Jan 2020-2021	Condensed Matter Field Theory	Kedar Damle
Jan-July. 2021	Statistical Physics II	Tridib Sadhu
Jan-July. 2023	Quantum Mechanics II	Vikram Tripathi

Publications

- **Melting temperature shifts from quantum fluctuations in generalized Wigner crystals**, [Aman Kumar](#), *Sogoud Sherif, Veit Elser and Hitesh J. Changlani*, [arXiv:2604.19870](#)
- **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, [Aman Kumar](#), *Cyprian Lewandowski, Hitesh Changlani*, [npj Quantum Materials 10, 95 \(2025\)](#)
- **Haerter-Shastry kinetic magnetism and metallicity in the triangular Hubbard model**, *Sogoud Sherif, Prakash Sharma*, [Aman Kumar](#), *Hitesh J. Changlani*, [arXiv:2510.18954](#)
- **Identifying chiral topological order in microscopic spin models by modular commutator**, *Avijit Maity*, [Aman Kumar](#), *Vikram Tripathi*, [arXiv:2510.06086](#)
- **Field tuning Kitaev systems for spin fractionalization and topological order**, *Jagannath Das, Sarbajaya Kundu*, [Aman Kumar](#), *Vikram Tripathi*, [J. Physics Condens. Matter 36 443001\(2024\)](#)
- **Thermal Hall conductivity near field-suppressed magnetic order in a Kitaev-Heisenberg model**, [Aman Kumar](#), *Vikram Tripathi*, [Phys. Rev. B 107, L220406 \(2023\)](#)
- **Quasiparticle metamorphosis in the random t - J model**, [Aman Kumar](#), *Subir Sachdev, Vikram Tripathi*, [Phys. Rev. B 106, L081120 \(2022\)](#)
- **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, [Aman Kumar](#), *Vikram Tripathi*, [Phys. Rev. B 102, 100401\(R\) \(2020\)](#)
- **Jordan-Wigner fermionization of quantum spin systems on arbitrary two-dimensional lattices: A mutual Chern-Simons approach**, *Jagannath Das*, [Aman Kumar](#), *Avijit Maity, and Vikram Tripathi*, [Phys. Rev. B 108, 085110 \(2023\)](#)
- **Microscopic theory of field-tuned topological transitions in the Kitaev honeycomb model**, *Jagannath Das*, [Aman Kumar](#), *and Vikram Tripathi*, [Phys. Rev. B 111, L041111 \(2025\)](#)

Conferences and Talks

- Mar 19, 2026 **Quantum and classical thermal melting of generalized Wigner crystals**, *APS March Meeting*
- Aug 29, 2025 **Finite-temperature study of generalized Wigner crystals**, *Random Interaction Seminar, TIFR Mumbai*
- Mar 19, 2025 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *APS March Meeting*

- Aug 21, 2024 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Seminar at IIT Bombay*
- Aug 8, 2024 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Random Interaction Seminar, TIFR Mumbai*
- Oct 7, 2022 **Thermal Hall conductivity near field-suppressed magnetic order in a Kitaev-Heisenberg model**, *Harvard University*
- Dec 11, 2021 **Quasiparticle stability in the hole-doped random t - j model**, *QMAT National Conference on Quantum Condensed Matter*

Posters Presented

- Aug 2–14, 2025 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *Workshop on Advances in Quantum Matter*
ICTP, Trieste, Italy
- Jan, 6–10, 2025 **Origin and stability of generalized Wigner crystallinity in triangular moiré systems**, *2025 Theory Winter School, Maglab*
NHFML, Tallahassee, Florida
- Dec 3–10, 2023 **Thermal Hall conductivity near field-suppressed magnetic order in a Kitaev-Heisenberg model**, *Conference on Fractionalization and Emergent Gauge Fields in Quantum Matter*
ICTP, Trieste, Italy
- Jun 20–25, 2022 **Quasiparticle metamorphosis in the hole-doped random t - j model**, *International Conference on Highly Frustrated Magnetism (HFM)*
Paris, France
- Apr 15–20, 2021 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Korrelationstage 2021 (online)*
- Feb 17–28, 2020 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Gapless Fermions – from Fermi liquids to strange metals*
Max Planck Institute for the Physics of Complex Systems (MPIPKS), Dresden (online)
- Dec 23, 2019 – Jan 2, 2020 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Novel Phases of Quantum Matter*
ICTS Bangalore, India
- Nov 11–29, 2019 **Kitaev quasiparticles in a proximate spin liquid: A many-body localization perspective**, *Thermalization, Many-body Localization, and Hydrodynamics*
ICTS Bangalore, India

References

Prof. Vikram Tripathi

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Prof. Hitesh Changlani

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Prof. Subir Sachdev

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Prof. Veit Elser

Department of Physics,

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